Now using the example 7.35, we have

$$\lim_{\theta\to 0}\frac{m}{n}\times\left(\frac{\frac{\sin m\theta}{\theta}}{\frac{\sin n\theta}{\theta}}\right)=\frac{m^2}{n^2}$$

$$\text{Therefore } m^2 = n^2$$

$$\text{That is } m = \pm n\,.$$

Example 7.38

Evaluate : $\lim_{x\to 1^-}\left(\frac{\log(1-x)}{\cot(\pi x)}\right)$.

Solution

This is an indeterminate form $\frac{\infty}{\infty}$ and hence we use the l'Hôpital's Rule to evaluate the limit.

$$\text{Thus, } \lim_{x\to 1^-}\frac{\log(1-x)}{\cot(\pi x)} = \lim_{x\to 1^-}\left(\frac{-\frac{1}{1-x}}{-\pi\operatorname{cosec}^2(\pi x)}\right)\left(\frac{\infty}{\infty}\text{ form}\right)$$

On simplication, we get

$$= \lim_{x\to 1^-}\left(\frac{\sin^2(\pi x)}{\pi(1-x)}\right).\quad\left(\frac{0}{0}\text{ form}\right)$$

again applying the l'Hôpital Rule, we get

$$= \lim_{x\to 1^-}\left(\frac{2\pi\sin(\pi x)\cdot\cos(\pi x)}{-\pi}\right)$$

$$= \lim_{x\to 1^-}\left(-2\sin(\pi x)\cdot\cos(\pi x)\right)$$

$$= 0\,.$$

Example 7.39

Evaluate : $\lim_{x\to 0^+}\left(\frac{1}{x}-\frac{1}{e^x-1}\right)$.

Solution

This is an indeterminate of the form $\infty-\infty$. To evaluate this limit we first simplify and bring it in the form $\left(\frac{0}{0}\right)$ and applying the l'Hôpital Rule, we get

$$\lim_{x\to 0^+}\left(\frac{1}{x}-\frac{1}{e^x-1}\right) = \lim_{x\to 0^+}\left(\frac{e^x-x-1}{x(e^x-1)}\right).\quad\left(\frac{0}{0}\text{ form}\right)$$

Now,

$$\lim_{x\to 0^+}\left(\frac{e^x-x-1}{x(e^x-1)}\right) = \lim_{x\to 0^+}\left(\frac{e^x-1}{xe^x+e^x-1}\right)\left(\frac{0}{0}\text{ form}\right)$$

$$= \lim_{x\to 0^+}\left(\frac{e^x}{xe^x+2e^x}\right)=\frac{1}{2}\,.$$

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